

341. If $a + b + c = 13$, $a^2 + b^2 + c^2 = 69$, then find $ab + bc + ca$.

- (a) -50 (b) 50
(c) 69 (d) 75

342. If $\frac{x^2 + y^2 + z^2 - 64}{xy - yz - zx} = -2$ and $x + y = 3z$, then the value of z is

- (a) 2 (b) 3
(c) 4 (d) None of these

343. $\left(\frac{785 \times 785 \times 785 + 435 \times 435 \times 435}{785 \times 785 + 435 \times 435 - 785 \times 435} \right)$ simplifies to

- (a) 350 (b) 785
(c) 1220 (d) 1320

344. $\left(\frac{147 \times 147 + 147 \times 143 + 143 \times 143}{147 \times 147 \times 147 - 143 \times 143 \times 143} \right) = ?$

- (a) $\frac{1}{4}$ (b) 290
(c) $\frac{1}{290}$ (d) 4

345. $\frac{(13)^3 + 7^3}{(13)^2 + 7^2 - ?} = 20$

- (a) 6 (b) 20
(c) 91 (d) None of these

346. The value of $\left(\frac{\left(\frac{3}{5} \right)^3 - \left(\frac{2}{5} \right)^3}{\left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2} \right)$ is (S.S.C., 2003)

- (a) $\frac{1}{5}$ (b) $\frac{19}{25}$
(c) $\frac{21}{25}$ (d) 1

347. $\frac{38 \times 38 \times 38 + 34 \times 34 \times 34 + 28 \times 28 \times 28 - 38 \times 34 \times 84}{38 \times 38 + 34 \times 34 + 28 \times 28 - 38 \times 34 - 34 \times 28 - 38 \times 28}$

is equal to

- (a) 24 (b) 32
(c) 44 (d) 100

348. The value of $\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{9(x-y)(y-z)(z-x)}$ is equal to

- (a) 0 (b) $\frac{1}{9}$
(c) $\frac{1}{3}$ (d) 1

349. If $x = 5$, $y = 3$, the value of $\frac{x^3 - y^3}{x^2 - y^2} - \frac{3xy}{x + y}$ will be

(P.C.S., 2009)

- (a) 1 (b) $\frac{1}{2}$
(c) $\frac{3}{2}$ (d) $\frac{5}{2}$

350. If $a + b + c = 11$ and $ab + bc + ca = 20$, then the value of the expression $a^3 + b^3 + c^3 - 3abc$ will be

- (a) 121 (b) 341
(c) 671 (d) 781

351. The value of $\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3}$ is

- (a) 0
(b) 1
(c) $[2(x+y+z)]^{-1}$
(d) $[(x+y)(y+z)(z+x)]^{-1}$

352. If $a + b + c = 0$, the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$ is

- (a) $3abc$ (b) $\frac{1}{3}$
(c) 1 (d) 3

353. If $a = 29$, $b = 24$, $c = 27$, the value of $a^3 + b^3 + c^3 - 3abc$ is

- (a) 1420 (b) 1520
(c) 1620 (d) 1920

354. If $\frac{x^2 - 1}{x} = 4$, then $\frac{x^6 - 1}{x^3}$ is

- (a) 63 (b) 66
(c) 75 (d) 76

355. $(32)^3 + (79)^3 - (111)^3 + 3 \times 32 \times 79 \times 111$ is equal to

- (a) 0 (b) 1
(c) 10000 (d) 30007

356. If $(x + y) = 3$, $xy = 2$, then what is the value of $x^3 + y^3$?

- (a) 6 (b) 7
(c) 8 (d) 9

(e) None of these

357. The value of $\frac{x^2 - (y-z)^2}{(x+z)^2 - y^2} + \frac{y^2 - (x-z)^2}{(x+y)^2 - z^2} + \frac{z^2 - (x-y)^2}{(y+z)^2 - x^2}$ is

(M.B.A., 2006)

- (a) -1 (b) 0
(c) 1 (d) None of these

358. If $\frac{p}{a} + \frac{q}{b} + \frac{r}{c} = 1$ and $\frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 0$ where $a, b, c, p,$

q, r are non-zero real numbers, then $\frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2}$ is

equal to

- (a) 0 (b) 1
(c) 3 (d) 9